1 $\frac{5-(-3)}{6-2}$ or equivalent **[M1]**

2 **[A1]**

[Total Marks: 2]

2 **(a)** Evidence of using $\frac{\text{change in } y}{\text{change in } x}$ with two correct points from the graph, e.g., $\frac{4-(-2)}{3-0}$ **[M1]**

2 **[A1]**

(b) $(0, -2)$ **[B1]**

[Total Marks: 3]

3 **(a)** Uses gradient formula: $\frac{-4-8}{4-(-2)}$ or $\frac{8-(-4)}{-2-4}$ **[M1]**

-2 **[A1]**

(b) Identifies coordinates of two clear points on the line, e.g., $(0, 1)$ and $(4, 4)$ **[M1]**

$\frac{3}{4}$ or 0.75 **[A1]**

[Total Marks: 4]

4 **(a)** Uses the gradient formula: $\frac{-5-7}{2-(-4)}$ **[M1]**

-2 **[A1]**

(b) Sets up the equation: $\frac{c-7}{0-(-4)} = -2$ **[M1]**

-1 **[A1]**

[Total Marks: 4]

5 **(a)** Correct vertical and horizontal change seen, e.g. $\frac{6}{4}$ or $\frac{3}{2}$ **[M1]**

1.5 (or $\frac{3}{2}$) **[A1]**

(b) Correct vertical and horizontal change seen, e.g. $\frac{-3}{6}$ **[M1]**

-0.5 (or $-\frac{1}{2}$) **[A1]**

(c) $\frac{-9-7}{5-(-3)}$ **[M1]**

-2 **[A1]**

[Total Marks: 6]

6 **(a)** $\dfrac{k-5}{4-(-2)} = -\dfrac{1}{2}$ **[M1]**

$k - 5 = -3$ **[M1]**

2 **[A1]**

(b) Gradient of $BC = \dfrac{-1-2}{10-4} = -\dfrac{1}{2}$ **[M1]**

Gradient of AB = gradient of BC and point B is common to both segments **[A1]**

(c) $\dfrac{(3p-1)-5}{p-(-2)} = 2$ **[M1]**

$3p - 6 = 2(p+2)$ **[M1]**

10 **[A1]**

[Total Marks: 8]

7 **(a)** $\dfrac{(a+4)-3a}{5a-2a}$ or equivalent **[M1]**

$\dfrac{4-2a}{3a}$ **[A1]**

(b) States or implies gradient of PQ is $-\dfrac{1}{3}$ **[B1]**

$\dfrac{4-2a}{3a} = -\dfrac{1}{3}$ or $3(4-2a) = -3a$ **[M1]**

4 **[A1]**

(c) Gradient $AB = \dfrac{8-2}{3-(-1)}$ or $\dfrac{6}{4}$ or $\dfrac{3}{2}$ **[M1]**

$\dfrac{23-8}{c-3} = \dfrac{3}{2}$ or $\dfrac{23-2}{c-(-1)} = \dfrac{3}{2}$ **[M1]**

13 **[A1]**

[Total Marks: 8]

8 **(a)** $\dfrac{(a+6)-3a}{4-(-2)}$ **[M1]**

$\dfrac{6-2a}{6}$ **[M1]**

$\dfrac{3-a}{3}$ or $1-\dfrac{a}{3}$ **[A1]**

(b) $\dfrac{3-a}{3} = 3$ oe **[M1]**

$a = -6$ **[A1]**

(c) $P = (-2,-18)$ and $Q = (4,0)$ **[M1]**

$\left(\dfrac{-2+4}{2}, \dfrac{-18+0}{2}\right)$ **[M1]**

$(1,-9)$ **[A1]**

[Total Marks: 8]

9 **(a)** $\frac{k-2}{4}$ **[B1]**

(b) Finds an expression for the gradient of BC (e.g., $\frac{k}{k-5}$) and equates to the gradient of AB: $\frac{k-2}{4} = \frac{k}{k-5}$ **[M1]**

Expands brackets correctly: $(k-2)(k-5) = 4k$

$\mathrm{k}^2 - 7k + 10 = 4k$ [M1]

Rearranges to the given equation $k^2 - 11k + 10 = 0$ with no mathematical errors seen **[A1]**

(c) Factorises $(k-1)(k-10) = 0$ or uses the quadratic formula **[M1]**

1 and 10 **[A1]**

(d) Deduces that $k = 10$ must be used, as $k = 1$ would mean points A and C are the same, contradicting that they are distinct **[B1]**

Calculates the gradient of OB using $(5, 10)$: $\frac{10-0}{5-0} = 2$ **[B1]**

[Total Marks: 8]

10 **(a)** Calculates the gradient of AB as $\frac{9-3}{4-1} = 2$ and equates to gradient of BC or AC, e.g., $\frac{y-9}{8-4} = 2$ **[M1]**

$y = 17$ **[A1]**

(b) Sets up the gradient equation: $\frac{(p^2-p-3)-3}{p-1} = 2$ **[M1]**

Simplifies to form a 3-term quadratic equation: $p^2 - p - 6 = 2(p-1)$

$\mathrm{p}^2 - 3p - 4 = 0$ [M1]

Factorises $(p-4)(p+1) = 0$ or uses the quadratic formula correctly **[M1]**

$p = 4$ and $p = -1$ **[A1]**

(c) Sets up the gradient equation: $\frac{12-m}{(m+1)-3} = m$ **[M1]**

Multiplies by the denominator: $12 - m = m(m-2)$ **[M1]**

Rearranges into a 3-term quadratic: $m^2 - m - 12 = 0$ **[M1]**

$m = 4$ and $m = -3$ **[A1]**

[Total Marks: 10]

11 **(a)** $\frac{5-1}{4-(-2)}$ or equivalent **[M1]**

$\frac{2}{3}$ **[A1]**

(b) $\dfrac{-2-k}{8-3} = -1.4$ or equivalent **[M1]**

$-2-k=-7$ **[M1]**

5 **[A1]**

[Total Marks: 5]

12 (a) $\dfrac{-9-5}{4-(-3)}$ or $\dfrac{5-(-9)}{-3-4}$ **[M1]**

-2 **[A1]**

(b) $\dfrac{1-5}{k-(-3)} = -2$ oe **[M1]**

$-4=-2(k+3)$ or $-2k-6=-4$ **[M1]**

$k=-1$ **[A1]**

[Total Marks: 5]

13 (a) Forms a correct equation for the gradient: $\dfrac{5-2k}{2-(k-3)} = -3$ OR $\dfrac{2k-5}{k-3-2} = -3$ **[M1]**

Correctly simplifies the denominator and equates: $5-2k=-3(5-k)$ **[M1]**

$k=4$ **[A1]**

(b) Evaluates coordinates of P as $(1,8)$ and substitutes into midpoint formula: $\left(\dfrac{1+2}{2}, \dfrac{8+5}{2}\right)$ **[M1]**

$(1.5, 6.5)$ **[A1]**

[Total Marks: 5]

14 (a) Finds a correct algebraic expression for the gradient: $\dfrac{8-4}{k-2-3k}$ or $\dfrac{4}{-2k-2}$ **[M1]**

Equates to $-\dfrac{1}{2}$ and eliminates fractions: $8=2k+2$ or $-8=-2k-2$ **[M1]**

3 **[A1]**

(b) $P(9,4)$ and $Q(1,8)$ **[B1]**

(c) Identifies the coordinates of R must be of the form $(0,y)$ **[B1]**

Sets up the gradient equation: $\dfrac{y-4}{0-9} = 1$ **[M1]**

$(0,-5)$ **[A1]**

[Total Marks: 7]

15 (a) Gradient $= \dfrac{(t^2-3t+2)-0}{t-1}$ **[M1]**

Factorising numerator: $\dfrac{(t-1)(t-2)}{t-1}$ **[M1]**

Cancelling $(t-1)$ correctly to obtain $t-2$ **[A1]**

(b) $t-2=2.8$ giving $t=4.8$ **[M1]**

$y=4.8^2-3(4.8)+2$ **[M1]**

$(4.8, 10.64)$ **[A1]**

(c) Gradient of $QR=\dfrac{(t^2-3t+2)-(-1.5)}{t-3.2}$ **[M1]**

$\dfrac{t^2-3t+3.5}{t-3.2}=t-2$ **[M1]**

$t^2-3t+3.5=t^2-5.2t+6.4$ leading to $2.2t=2.9$ **[M1]**

1.32 **[A1]**

[Total Marks: 10]